and the two data values 62 and 67 can be represented as

Stem	Leaf
6	2,7

Example 2.2.c. Table 2.5 gives the monthly and yearly average daily minimum temperatures in 35 U.S. cities.

The annual average daily minimum temperatures from Table 2.5 are represented in the following stem and leaf plot.

7 0.0
6 9.0
5 1.0, 1.3, 2.0, 5.5, 7.1, 7.4, 7.6, 8.5, 9.3
4 0.0, 1.0, 2.4, 3.6, 3.7, 4.8, 5.0, 5.2, 6.0, 6.7, 8.1, 9.0, 9.2
3 3.1, 4.1, 5.3, 5.8, 6.2, 9.0, 9.5, 9.5
2 9.0, 9.8 ■

2.3 Summarizing data sets

Modern-day experiments often deal with huge sets of data. For instance, in an attempt to learn about the health consequences of certain common practices, in 1951 the medical statisticians R. Doll and A. B. Hill sent questionnaires to all doctors in the United Kingdom and received approximately 40,000 replies. Their questions dealt with age, eating habits, and smoking habits. The respondents were then tracked for the ensuing 10 years and the causes of death for those who died were monitored. To obtain a feel for such a large amount of data, it is useful to be able to summarize it by some suitably chosen measures. In this section we present some summarizing *statistics*, where a statistic is a numerical quantity whose value is determined by the data.

2.3.1 Sample mean, sample median, and sample mode

In this section we introduce some statistics that are used for describing the center of a set of data values. To begin, suppose that we have a data set consisting of the n numerical values $x_1, x_2, \ldots, x_n$. The sample mean is the arithmetic average of these values.

Definition. The *sample mean*, designated by $\bar{x}$, is defined by

$$\bar{x} = \sum_{i=1}^{n} x_i/n$$

The computation of the sample mean can often be simplified by noting that if for constants a and b

$$y_i = ax_i + b, \qquad i = 1, \ldots, n$$

Table 2.5 Normal Daily Minimum Temperature — Selected Cities
In Fahrenheit degrees. Airport data except as noted. Based on standard 30-year period, 1961 through 1990

State	Station	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sept	Oct.	Nov.	Dec.	Annual avg.
AL	Mobile	40.0	42.7	50.1	57.1	64.4	70.7	73.2	72.9	68.7	57.3	49.1	43.1	57.4
AK	Juneau	19.0	22.7	26.7	32.1	38.9	45.0	48.1	47.3	42.9	37.2	27.2	22.6	34.1
AZ	Phoenix	41.2	44.7	48.8	55.3	63.9	72.9	81.0	79.2	72.8	60.8	48.9	41.8	59.3
AR	Little Rock	29.1	33.2	42.2	50.7	59.0	67.4	71.5	69.8	63.5	50.9	41.5	33.1	51.0
CA	Los Angeles	47.8	49.3	50.5	52.8	56.3	59.5	62.8	64.2	63.2	59.2	52.8	47.9	55.5
	Sacramento	37.7	41.4	43.2	45.5	50.3	55.3	58.1	58.0	55.7	50.4	43.4	37.8	48.1
	San Diego	48.9	50.7	52.8	55.6	59.1	61.9	65.7	67.3	65.6	60.9	53.9	48.8	57.6
	San Francisco	41.8	45.0	45.8	47.2	49.7	52.6	53.9	55.0	55.2	51.8	47.1	42.7	49.0
CO	Denver	16.1	20.2	25.8	34.5	43.6	52.4	58.6	56.9	47.6	36.4	25.4	17.4	36.2
CT	Hartford	15.8	18.6	28.1	37.5	47.6	56.9	62.2	60.4	51.8	40.7	32.8	21.3	39.5
DE	Wilmington	22.4	24.8	33.1	41.8	52.2	61.6	67.1	65.9	58.2	45.7	37.0	27.6	44.8
DC	Washington	26.8	29.1	37.7	46.4	56.6	66.5	71.4	70.0	62.5	50.3	41.1	31.7	49.2
FL	Jacksonville	40.5	43.3	49.2	54.9	62.1	69.1	71.9	71.8	69.0	59.3	50.2	43.4	57.1
	Miami	59.2	60.4	64.2	67.8	72.1	75.1	76.2	76.7	75.9	72.1	66.7	61.5	69.0
GA	Atlanta	31.5	34.5	42.5	50.2	58.7	66.2	69.5	69.0	63.5	51.9	42.8	35.0	51.3
HI	Honolulu	65.6	65.4	67.2	68.7	70.3	72.2	73.5	74.2	73.5	72.3	70.3	67.0	70.0
ID	Boise	21.6	27.5	31.9	36.7	43.9	52.1	57.7	56.8	48.2	39.0	31.1	22.5	39.1
IL	Chicago	12.9	17.2	28.5	38.6	47.7	57.5	62.6	61.6	53.9	42.2	31.6	19.1	39.5
	Peoria	13.2	17.7	29.8	40.8	50.9	60.7	65.4	63.1	55.2	43.1	32.5	19.3	41.0
IN	Indianapolis	17.2	20.9	31.9	41.5	51.7	61.0	65.2	62.8	55.6	43.5	34.1	23.2	42.4
IA	Des Moines	10.7	15.6	27.6	40.0	51.5	61.2	66.5	63.6	54.5	42.7	29.9	16.1	40.0
KS	Wichita	19.2	23.7	33.6	44.5	54.3	64.6	69.9	67.9	59.2	46.6	33.9	23.0	45.0
KY	Louisville	23.2	26.5	36.2	45.4	54.7	62.9	67.3	65.8	58.7	45.8	37.3	28.6	46.0
LA	New Orleans	41.8	44.4	51.6	58.4	65.2	70.8	73.1	72.8	69.5	58.7	51.0	44.8	58.5
ME	Portland	11.4	13.5	24.5	34.1	43.4	52.1	58.3	57.1	48.9	38.3	30.4	17.8	35.8
MD	Baltimore	23.4	25.9	34.1	42.5	52.6	61.8	66.8	65.7	58.4	45.9	37.1	28.2	45.2
MA	Boston	21.6	23.0	31.3	40.2	49.8	59.1	65.1	64.0	56.8	46.9	38.3	26.7	43.6
MI	Detroit	15.6	17.6	27.0	36.8	47.1	56.3	61.3	59.6	52.5	40.9	32.2	21.4	39.0
	Sault Ste. Marie	4.6	4.8	15.3	28.4	38.4	45.5	51.3	51.3	44.3	36.2	25.9	11.8	29.8
MN	Duluth	−2.2	2.8	15.7	28.9	39.6	48.5	55.1	53.3	44.5	35.1	21.5	4.9	29.0
	Minneapolis-St. Paul	2.8	9.2	22.7	36.2	47.6	57.6	63.1	60.3	50.3	38.8	25.2	10.2	35.3
MS	Jackson	32.7	35.7	44.1	51.9	60.0	67.1	70.5	69.7	63.7	50.3	42.3	36.1	52.0
MO	Kansas City	16.7	21.8	32.6	43.8	53.9	63.1	68.2	65.7	56.9	45.7	33.6	21.9	43.7
	St. Louis	20.8	25.1	35.5	46.4	56.0	65.7	70.4	67.9	60.5	48.3	37.7	26.0	46.7
MT	Great Falls	11.6	17.2	22.8	31.9	40.9	48.6	53.2	52.2	43.5	35.8	24.3	14.6	33.1

Source: U.S. National Oceanic and Atmospheric Administration, Climatography of the United States, No. 81.

then the sample mean of the data set $y_1, \ldots, y_n$ is

$$\bar{y} = \sum_{i=1}^{n} (ax_i + b)/n = \sum_{i=1}^{n} ax_i/n + \sum_{i=1}^{n} b/n = a\bar{x} + b$$

Example 2.3.a. The winning scores in the U.S. Masters golf tournament in the years from 2004 to 2013 were as follows:

280, 278, 272, 276, 281, 279, 276, 281, 289, 280

Find the sample mean of these scores.

Solution. Rather than directly adding these values, it is easier to first subtract 280 from each one to obtain the new values $y_i = x_i - 280$:

$$0, -2, -8, -4, 1, -1, -4, 1, 9, 0$$

Because the arithmetic average of the transformed data set is

$$\bar{y} = -8/10$$

it follows that

$$\bar{x} = \bar{y} + 280 = 279.2 \quad \blacksquare$$

Sometimes we want to determine the sample mean of a data set that is presented in a frequency table listing the k distinct values $v_1, \ldots, v_k$ having corresponding frequencies $f_1, \ldots, f_k$. Since such a data set consists of $n = \sum_{i=1}^{k} f_i$ observations, with the value v_i appearing f_i times, for each $i = 1, \ldots, k$, it follows that the sample mean of these n data values is

$$\bar{x} = \sum_{i=1}^{k} v_i f_i / n$$

By writing the preceding as

$$\bar{x} = \frac{f_1}{n} v_1 + \frac{f_2}{n} v_2 + \cdots + \frac{f_k}{n} v_k$$

we see that the sample mean is a *weighted average* of the distinct values, where the weight given to the value v_i is equal to the proportion of the n data values that are equal to $v_i, i = 1, \ldots, k$.

Example 2.3.b. The following is a frequency table giving the ages of members of a symphony orchestra for young adults.

Age	Frequency
15	2
16	5
17	11
18	9
19	14
20	13

Find the sample mean of the ages of the 54 members of the symphony.

Solution.

$$\bar{x} = (15 \cdot 2 + 16 \cdot 5 + 17 \cdot 11 + 18 \cdot 9 + 19 \cdot 14 + 20 \cdot 13)/54 \approx 18.24 \quad \blacksquare$$

Another statistic used to indicate the center of a data set is the *sample median*; loosely speaking, it is the middle value when the data set is arranged in increasing order.

Definition. Order the values of a data set of size n from smallest to largest. If n is odd, the *sample median* is the value in position $(n + 1)/2$; if n is even, it is the average of the values in positions $n/2$ and $n/2 + 1$.

Thus the sample median of a set of three values is the second smallest; of a set of four values, it is the average of the second and third smallest.

Example 2.3.c. Find the sample median for the data described in Example 2.3.b.

Solution. Since there are 54 data values, it follows that when the data are put in increasing order, the sample median is the average of the values in positions 27 and 28. Thus, the sample median is 18.5. $\blacksquare$

The sample mean and sample median are both useful statistics for describing the central tendency of a data set. The sample mean makes use of all the data values and is affected by extreme values that are much larger or smaller than the others; the sample median makes use of only one or two of the middle values and is thus not affected by extreme values. Which of them is more useful depends on what one is trying to learn from the data. For instance, if a city government has a flat rate income tax and is trying to estimate its total revenue from the tax, then the sample mean of its residents' income would be a more useful statistic. On the other hand, if the city was thinking about constructing middle-income housing, and wanted to determine the proportion of its population able to afford it, then the sample median would probably be more useful.

Example 2.3.d. In a study reported in Hoel, D. G., "A representation of mortality data by competing risks," *Biometrics*, **28**, pp. 475–488, 1972, a group of 5-week-old mice were each given a radiation dose of 300 rad. The mice were then divided into two groups; the first group was kept in a germ-free environment, and the second in conventional laboratory conditions. The numbers of days until death were then observed. The data for those whose death was due to thymic lymphoma are given in the following stem and leaf plots (whose stems are in units of hundreds of days); the first plot is for mice living in the germ-free conditions and the second for mice living under ordinary laboratory conditions.

Germ-Free Mice

1	58, 92, 93, 94, 95
2	02, 12, 15, 29, 30, 37, 40, 44, 47, 59
3	01, 01, 21, 37
4	15, 34, 44, 85, 96
5	29, 37
6	24
7	07
8	00

Conventional Mice

1	59, 89, 91, 98
2	35, 45, 50, 56, 61, 65, 66, 80
3	43, 56, 83
4	03, 14, 28, 32

Determine the sample means and the sample medians for the two sets of mice.

Solution. It is clear from the stem and leaf plots that the sample mean for the set of mice put in the germ-free setting is larger than the sample mean for the set of mice in the usual laboratory setting; indeed, a calculation gives that the former sample mean is 344.07, whereas the latter one is 292.32. On the other hand, since there are 29 data values for the germ-free mice, the sample median is the 15th largest data value, namely, 259; similarly, the sample median for the other set of mice is the 10th largest data value, namely, 265. Thus, whereas the sample mean is quite a bit larger for the first data set, the sample medians are approximately equal. The reason for this is that whereas the sample mean for the first set is greatly affected by the five data values greater than 500, these values have a much smaller effect on the sample median. Indeed, the sample median would remain unchanged if these values were replaced by any other five values greater than or equal to 259. It appears from the stem and leaf plots that the germ-free conditions probably improved the life span of the five longest living rats, but it is unclear what, if any, effect it had on the life spans of the other rats. ■

Another statistic that has been used to indicate the central tendency of a data set is the *sample mode*, defined to be the value that occurs with the greatest frequency. If no single value occurs most frequently, then all the values that occur at the highest frequency are called *modal values*.

Example 2.3.e. The following frequency table gives the values obtained in 40 rolls of a die.

Value	Frequency
1	9
2	8
3	5
4	5
5	6
6	7

Find **(a)** the sample mean, **(b)** the sample median, and **(c)** the sample mode.

Solution. **(a)** The sample mean is

$$\bar{x} = (9 + 16 + 15 + 20 + 30 + 42)/40 = 3.05$$

(b) The sample median is the average of the 20th and 21st smallest values, and is thus equal to 3. **(c)** The sample mode is 1, the value that occurred most frequently. ■

2.3.2 Sample variance and sample standard deviation

Whereas we have presented statistics that describe the central tendencies of a data set, we are also interested in ones that describe the spread or variability of the data values. A statistic that could be used for this purpose would be one that measures the average value of the squares of the distances between the data values and the sample mean. This is accomplished by the sample variance, which for technical reasons divides the sum of the squares of the differences by $n - 1$ rather than n, where n is the size of the data set.

Definition. The *sample variance*, call it s^2, of the data set $x_1, \ldots, x_n$ is defined by

$$s^2 = \sum_{i=1}^{n} (x_i - \bar{x})^2/(n-1)$$

Example 2.3.f. Find the sample variances of the data sets **A** and **B** given below.

A: 3, 4, 6, 7, 10 **B:** −20, 5, 15, 24

Solution. As the sample mean for data set **A** is $\bar{x} = (3 + 4 + 6 + 7 + 10)/5 = 6$, it follows that its sample variance is

$$s^2 = [(-3)^2 + (-2)^2 + 0^2 + 1^2 + 4^2]/4 = 7.5$$

The sample mean for data set **B** is also 6; its sample variance is

$$s^2 = [(-26)^2 + (-1)^2 + 9^2 + (18)^2]/3 \approx 360.67$$

Thus, although both data sets have the same sample mean, there is a much greater variability in the values of the **B** set than in the **A** set. ■

The following algebraic identity is often useful for computing the sample variance:

An algebraic identity

$$\sum_{i=1}^{n}(x_i-\bar{x})^2=\sum_{i=1}^{n}x_i^2-n\bar{x}^2$$

The identity is proven as follows:

$$\begin{aligned}\sum_{i=1}^{n}(x_i-\bar{x})^2&=\sum_{i=1}^{n}\left(x_i^2-2x_i\bar{x}+\bar{x}^2\right)\\&=\sum_{i=1}^{n}x_i^2-2\bar{x}\sum_{i=1}^{n}x_i+\sum_{i=1}^{n}\bar{x}^2\\&=\sum_{i=1}^{n}x_i^2-2n\bar{x}^2+n\bar{x}^2\\&=\sum_{i=1}^{n}x_i^2-n\bar{x}^2\end{aligned}$$

The computation of the sample variance can also be eased by noting that if

$$y_i=a+bx_i,\qquad i=1,\ldots,n$$

then $\bar{y}=a+b\bar{x}$, and so

$$\sum_{i=1}^{n}(y_i-\bar{y})^2=b^2\sum_{i=1}^{n}(x_i-\bar{x})^2$$

That is, if s_y^2 and s_x^2 are the respective sample variances, then

$$s_y^2=b^2s_x^2$$

In other words, adding a constant to each data value does not change the sample variance; whereas multiplying each data value by a constant results in a new sample variance that is equal to the old one multiplied by the square of the constant. ■

Example 2.3.g. The following data give the worldwide number of fatal airline accidents of commercially scheduled air transports in the years from 1997 to 2005.

Year	1997	1998	1999	2000	2001	2002	2003	2004	2005
Accidents	25	20	21	18	13	13	7	9	18

Source: National Safety Council.

Find the sample variance of the number of accidents in these years.

Solution. Let us start by subtracting 18 from each value, to obtain the new data set:

$$7, 2, 3, 0, -5, -5, -11, -9, 0$$

Calling the transformed data $y_1, \ldots, y_9$, we have

$$\bar{y} = \sum_{i=1}^{9} y_i/9 = -2, \qquad \sum_{i=1}^{n} y_i^2 = 49 + 4 + 9 + 25 + 25 + 121 + 81 = 314$$

Hence, since the sample variance of the transformed data is equal to that of the original data, upon using the algebraic identity we obtain

$$s^2 = \frac{314 - 9(4)}{8} = 34.75 \quad \blacksquare$$

Program 2.3 on the text disk can be used to obtain the sample variance for large data sets.

The positive square root of the sample variance is called the *sample standard deviation*.

Definition. The quantity s, defined by

$$s = \sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2/(n-1)}$$

is called the *sample standard deviation*.

The sample standard deviation is measured in the same units as the data.

2.3.3 Sample percentiles and box plots

Loosely speaking, the sample $100p$ percentile of a data set is that value such that $100p$ percent of the data values are less than or equal to it, $0 \leq p \leq 1$. More formally, we have the following definition.

Definition. The *sample 100*p *percentile* is that data value such that at least $100p$ percent of the data are less than or equal to it and at least $100(1-p)$ percent are greater than or equal to it. If two data values satisfy this condition, then the sample $100p$ percentile is the arithmetic average of these two values.

To determine the sample $100p$ percentile of a data set of size n, we need to determine the data values such that

1. At least np of the values are less than or equal to it.
2. At least $n(1-p)$ of the values are greater than or equal to it.

To accomplish this, first arrange the data in increasing order. Then, note that if np is not an integer, then the only data value that satisfies the preceding conditions is the one whose position when the data are ordered from smallest to largest is the smallest integer exceeding np. For instance, if $n = 22,\ p = .8$, then we require a data value such that at least 17.6 of the values are less than or equal to it, and at least 4.4 of them are greater than or equal to it. Clearly, only the 18th smallest value satisfies both conditions and this is the sample 80 percentile. On the other hand, if np is an integer, then it is easy to check that both the values in positions np and $np+1$ satisfy the preceding conditions, and so the sample $100p$ percentile is the average of these values. For instance, if we wanted the 90 percentile of a data set of size 20, then both the (18)th and (19)th smallest values would be such that at least 90 percent of the data values are less than or equal to them, and at least 10 percent of the data values are greater than or equal to them. Thus, the 90 percentile is the average of these two values.

Example 2.3.h. Table 2.6 lists the populations of the 25 most populous U.S. cities for the year 2006. For this data set, find **(a)** the sample 10 percentile and **(b)** the sample 80 percentile.

Solution. **(a)** Because the sample size is 25 and $25(.10) = 2.5$, the sample 10 percentile is the third smallest value, equal to 590, 763.

(b) Because $25(.80) = 20$, the sample 80 percentile is the average of the twentieth and the twenty-first smallest values. Hence, the sample 80 percentile is

$$\frac{1{,}512{,}986 + 1{,}448{,}394}{2} = 1{,}480{,}690 \quad \blacksquare$$

The sample 50 percentile is, of course, just the sample median. Along with the sample 25 and 75 percentiles, it makes up the sample quartiles.

Definition. The sample 25 percentile is called the *first quartile*; the sample 50 percentile is called the sample median or the *second quartile*; the sample 75 percentile is called the *third quartile*.

The quartiles break up a data set into four parts, with roughly 25 percent of the data being less than the first quartile, 25 percent being between the first and second quartile, 25 percent being between the second and third quartile, and 25 percent being greater than the third quartile.

Table 2.6 Population of 25 Largest U.S. Cities, July 2006.

Rank	City	Population
1	New York, NY	8,250,567
2	Los Angeles, CA	3,849,378
3	Chicago, IL	2,833,321
4	Houston, TX	2,144,491
5	Phoenix, AR	1,512,986
6	Philadelphia, PA	1,448,394
7	San Antonio, TX	1,296,682
8	San Diego, CA	1,256,951
9	Dallas, TX	1,232,940
10	San Jose, CA	929, 936
11	Detroit, MI	918, 849
12	Jacksonville, FL	794, 555
13	Indianapolis, IN	785, 597
14	San Francisco, CA	744, 041
15	Columbus, OH	733, 203
16	Austin, TX	709, 893
17	Memphis, TN	670, 902
18	Fort Worth, TX	653, 320
19	Baltimore, MD	640, 961
20	Charlotte, NC	630, 478
21	El Paso, TX	609, 415
22	Milwaukee, WI	602, 782
23	Boston, MA	590, 763
24	Seattle, WA	582, 454
25	Washington, DC	581, 530

Example 2.3.i. Noise is measured in decibels, denoted as dB. One decibel is about the level of the weakest sound that can be heard in a quiet surrounding by someone with good hearing; a whisper measures about 30 dB; a human voice in normal conversation is about 70 dB; a loud radio is about 100 dB. Ear discomfort usually occurs at a noise level of about 120 dB.

The following data give noise levels measured at 36 different times directly outside of Grand Central Station in Manhattan.

82, 89, 94, 110, 74, 122, 112, 95, 100, 78, 65, 60, 90, 83, 87, 75, 114, 85
69, 94, 124, 115, 107, 88, 97, 74, 72, 68, 83, 91, 90, 102, 77, 125, 108, 65

Determine the quartiles.